

LLARS - Script Creator Library

Current Creator API Recipes

Welcome to the LLARS copy/paste reference for current script creators.

This library is a companion to the Script Creator & Framework Guide. The guide explains the framework model; this document focuses on current Creator API recipes you can paste and adapt.

Traditional IniRead/Random/Send/label-SetTimer patterns are intentionally moved to the separate Legacy Script Creator Guide so this library can stay clean and API-first.

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1. How to Use This Library

Most current LLARS scripts can be built from a few calls: perform an action, wait using a named Config.ini section, let LLARS own the lifecycle, and read configuration directly only when task logic needs a value.

```
Run(ctx)
{
    LLARS_SetStatus("Banking")
    LLARS_Click("Bank")
    LLARS_Sleep("Sleep Short")
    LLARS_PressHotkey("Bank Preset")
    LLARS_Sleep("Final Sleep", true)
}
```

Note: If you are about to write IniRead + Random + Click/Sleep for a normal configured action, stop and check whether LLARS_Click(), LLARS_Sleep(), LLARS_PressHotkey(), or a typed config helper already does it.

2. Config.ini Building Blocks

Area click

```
#Area the script should click.
[Action Location]
type=coordinate
xmin=
xmax=
ymin=
ymax=
```

Exact point

```
#Exact pixel the script should monitor.
[Pixel Coordinate]
type=coordinate
x=
y=
```

Hotkey

```
#RuneScape hotkey for this action.
[Action Hotkey]
type=hotkey
hotkey=
```

Color

```
#Color that represents the desired state.
[Target Color]
type=color
coordinate=Pixel Coordinate
color=0xFFFFFFFF
```

Sleep/timer

```
[Sleep Short]
min=500
max=1000

[Action Timer]
min=10000
max=30000
```

Optional typed setting

```
[Optional Hotkey]
option=false
type=hotkey
hotkey=
```

3. RunCount Recipes

Basic

```
Start:

LLARS_RunCount(Func("Run"))

return

Run(ctx)
{
    LLARS_Click("Action Location")
    LLARS_Sleep("Final Sleep", true)
}
```

Different first run

```
Run(ctx)
{
    if (ctx.IsFirst)
    {
        LLARS_Click("Prime Location")
    }
    else
    {
        LLARS_Click("Normal Location")
    }

    LLARS_Sleep("Final Sleep", true)
}
```

Last run special action

```
if (ctx.IsLast)
    LLARS_SetStatus("Finishing")
```

Progress-aware creator logic

```
Log("PROGRESS", "Run " . ctx.Iteration . "/" . ctx.Total)  
remaining := ctx.Remaining
```

4. Click Recipes

Left click configured area

```
LLARS_Click("Action Location")
```

Right click

```
LLARS_Click("Action Location", "right")
```

Two clicks with travel wait

```
LLARS_Click("First Location")  
LLARS_Sleep("Travel Time")  
LLARS_Click("Second Location")
```

Optional click

```
if LLARS_ConfigEnabled("Optional Click")  
    LLARS_Click("Optional Click")
```

LLARS_Click() handles fixed points and rectangular random ranges. NaturalClick performs the movement and will not physically click until the cursor reaches the exact selected target pixel.

5. Keyboard Recipes

Configured hotkey

```
LLARS_PressHotkey("Bank Preset")
```

Purposeful hardcoded key

```
LLARS_PressKey("Space")  
LLARS_PressKey("Enter")  
LLARS_PressKey("4")
```

Mouse wheel

```
LLARS_PressKey("WheelDown")
```

Click then fixed key

```
LLARS_Click("Action Location")
LLARS_Sleep("Sleep Short")
LLARS_PressKey("Space")
```

Note: Do not make a key configurable just because it is hardcoded. Use `LLARS_PressKey()` when the fixed key is purposeful script behavior; use `LLARS_PressHotkey()` when the user is expected to choose the binding.

6. Status Recipes

```
LLARS_SetStatus("Banking")
LLARS_SetStatus("Waiting", "Portable")
LLARS_SetStatus("Cutting", "Refined Plank")
```

Status is useful around longer sequences and makes Developer Mode and the main GUI easier to follow.

7. Sleep and Runtime Recipes

Ordinary configured wait

```
LLARS_Sleep("Sleep Short")
```

True final RunCount wait

```
LLARS_Sleep("Sleep Craft", true)
```

First/following timing

```
if (ctx.IsFirst)
{
    LLARS_Sleep("First Setup Sleep")
}
else
{
    LLARS_Sleep("Normal Travel Sleep")
}

LLARS_Sleep("Final Sleep", true)
```

- Keep these calls inside the exact editing markers.
- Use literal section names when the pre-run estimator needs to recognize them.
- The current-loop countdown becomes exact at the final=true sleep.

8. Random Sleep Recipes

Normal use

```
LLARS_RandomSleep( )
```

Know the roll before later logic

```
willSleep := LLARS_RandomSleepRoll()

if (willSleep)
    LLARS_SetStatus("Random Sleep Pending")

LLARS_RandomSleep()
```

LLARS_RandomSleep() consumes a prepared roll if one exists; otherwise it rolls when called. The RunCount estimate uses the probability-weighted expected contribution and adjusts the current loop to what actually happened.

9. Optional Feature Recipes

Simple boolean

```
[Feature]
option=false

if LLARS_ConfigReadBool("Feature")
{
    ; feature behavior
}
```

Optional configured hotkey

```
[Special Action]
option=false
type=hotkey
hotkey=

if LLARS_ConfigEnabled("Special Action")
    LLARS_PressHotkey("Special Action")
```

Dependent setting

```
[Cannon]
option=false

[Cannon Restock]
depends=Cannon
type=coordinate
xmin=
xmax=
ymin=
ymax=
```

10. Typed Config Read Recipes

Integer with minimum

```
count := LLARS_ConfigReadInteger("Settings", "count", "", 1)
```

Number with range

```
offset := LLARS_ConfigReadNumber("Settings", "offset", 0, -50, 50)
```

Required text

```
name := LLARS_ConfigReadString("Settings", "name", "", true)
```

Choice

```
mode := LLARS_ConfigReadChoice("Settings", "mode", "One|Two|Three", "One")
```

Resolved point

```
if LLARS_ConfigReadPoint("Action Location", x, y)
{
    ; task-specific calculations using x/y
}
```

Configured range

```
if LLARS_ConfigReadRange("Action Timer", minValue, maxValue)
{
    ; task-specific use of the range
}
```

11. One-Shot Timer Recipes

```
ActionTimerID := LLARS_TimerOnce("Action Timer", Func("Action"))

Action()
{
    LLARS_PressKey("Space")
}
```

A one-shot callback is removed after it fires. If the run ends before it fires, framework cleanup removes it.

12. Repeating Timer Recipes

```
ActionTimerID := LLARS_TimerRepeat("Action Timer", Func("Action"))

Action()
{
    LLARS_Click("Action Location")
}
```

Every repeat receives a newly randomized interval from the timer section. This is the preferred replacement for manual randomize/reschedule label timers.

Stop it early

```
LLARS_TimerStop(ActionTimerID)
```

13. Bound Callback Recipes

Use `Func().Bind()` when one callback should serve several configured actions.

```
ScheduleAction(section, statusText)
{
    return LLARS_TimerOnce(section, Func("ActionTimer").Bind(section, statusText))
}

ActionTimer(section, statusText)
{
    LLARS_SetStatus(statusText)
    LLARS_PressHotkey(section)
}
```

This pattern is useful in scripts such as combat helpers where many optional sections share the same scheduling logic.

14. Timer Script Recipe

The maintained Timer template owns `LLARS_StartTimerRun()`, the overall countdown, `EndMsg`, and completion. Creator code normally supplies setup plus callbacks:

```
TimerSetup()
{
    global ScriptTimerID
    ScriptTimerID := LLARS_TimerRepeat("Script Timer", Func("ScriptTimer"))
}

ScriptTimer()
{
    LLARS_SetStatus("Working")
    LLARS_Click("Action Location")
}
```

Note: Timer templates prompt for an overall duration. For task-defined completion, use the `UntilDone` lifecycle (`LLARS_SCRIPT_TYPE := "UntilDone"` plus start/end helpers); no duration or run-count prompt is used.

15. MultiTimer Recipe

```
MultiTimerSetup()
{
    global TimerA, TimerB, TimerC

    TimerA := LLARS_TimerRepeat("Timer A", Func("ActionA"))
    TimerB := LLARS_TimerRepeat("Timer B", Func("ActionB"))
    TimerC := LLARS_TimerRepeat("Timer C", Func("ActionC"))
}

ActionA()
{
    LLARS_PressHotkey("Hotkey A")
}
```

```

}

ActionB()
{
    LLARS_Click("Location B")
}

ActionC()
{
    LLARS_PressKey("Space")
}

```

16. Single Color Detection

```

CheckPixel()
{
    global CheckPixelID, ResetCheckID

    if LLARS_PixelMatches("Pixel Coordinate", "Target Color")
    {
        LLARS_TimerStop(CheckPixelID)
        LLARS_Sleep("Sleep Timer")

        if !LLARS_RunActive()
            return

        LLARS_Click("Action Location")
        ResetCheckID := LLARS_TimerRepeat("Pixel Check Timer", Func("ResetCheck"))
    }
}

```

The API handles the pixel read and focus safety. The callback only needs task-specific trigger/reset behavior.

17. Multi Color Detection

```

ColorSections := ["Target Color One", "Target Color Two", "Target Color Three"]
matched := LLARS_PixelMatchesAny("Pixel Coordinate", ColorSections)

if (matched != "")
{
    LLARS_SetStatus("Detected", matched)
    LLARS_Click("Action Location")
}

```

Rotate matching action locations

```

LocationSections := ["Action Location One", "Action Location Two"]
locationIndex := 1

locationSection := LocationSections[locationIndex]
LLARS_Click(locationSection)

locationIndex++
if (locationIndex > LocationSections.Length())
    locationIndex := 1

```

18. Prevent Repeated Pixel Triggers

After one match is handled, stop detection until the watched pixel leaves every configured target state.

```
ResetCheck()  
{  
    global CheckPixelID, ResetCheckID, ColorSections  
  
    if (LLARS_PixelMatchesAny("Pixel Coordinate", ColorSections) = "")  
    {  
        LLARS_TimerStop(ResetCheckID)  
        CheckPixelID := LLARS_TimerRepeat("Pixel Check Timer", Func("CheckPixel"))  
    }  
}
```

This keeps one unchanged screen state from producing repeated clicks.

19. Dynamic Coordinates and NaturalClick

Use LLARS_Click() for normal Config.ini coordinate sections. Use NaturalClick only when creator logic has already calculated the final destination.

```
x := baseX + offsetX  
y := baseY + offsetY  
NaturalClick(x, y)
```

NaturalClick caps each calculated movement path at 3000 ms, but has no fixed retry or total elapsed-time limit. If the user moves the cursor, LLARS keeps correcting and clicks only after reaching the exact target pixel.

Note: When Developer Mode is open and LLARS can identify the matching configured range, the active coordinate box is shown during the click and remains visible briefly after a successful click.

20. Custom Raw Game-Facing Actions

Sometimes a script needs an operation that has no dedicated Creator API wrapper, such as a specialized MouseMove. Restore RuneScape immediately before the raw physical action.

```
if !LLARS_WaitForRuneScape("Custom MouseMove")  
    return  
  
MouseMove, %x%, %y%, %speed%
```

If more time-consuming logic occurs before the physical operation, call LLARS_WaitForRuneScape() again immediately before dispatch. LLARS_Sleep() restores RuneScape when it finishes; raw Sleep does not.

21. Common Action Combinations

Bank -> preset -> action -> fixed key

```
LLARS_SetStatus("Banking")  
LLARS_Click("Bank")  
LLARS_Sleep("Sleep Short")  
LLARS_PressHotkey("Bank Preset")  
LLARS_Sleep("Sleep Short")
```

```
LLARS_Click("Action Location")
LLARS_Sleep("Sleep Travel")
LLARS_PressKey("Space")
```

First run prime, later runs normal

```
if (ctx.IsFirst)
    LLARS_Click("Bank Prime")
else
    LLARS_Click("Bank Main")

LLARS_PressHotkey("Bank Preset")
LLARS_Sleep("Final Sleep", true)
```

Optional timed hotkey

```
if LLARS_ConfigReadBool("Potion")
    PotionTimerID := LLARS_TimerRepeat("Potion", Func("PotionAction"))

PotionAction()
{
    LLARS_PressHotkey("Potion")
}
```

22. Developer Mode Testing Recipes

- Open Developer Mode before a click and confirm the coordinate range appears during the click and briefly after success.
- Move the mouse while NaturalClick is traveling and confirm it corrects without clicking early.
- For multi-color scripts, scroll Live Inspector and compare Current Pixel to every Pixel Target.
- For Timer, confirm Progress matches the main GUI; for UntilDone, verify its completion condition ends the run.
- Scroll Active Hotkeys and Live Inspector; both should preserve the user position while data refreshes.
- Switch to Lightweight Mode and confirm the Developer Mode hotkey toggles Lightweight until Full Mode is selected again.
- Alt-Tab during LLARS_Sleep() and confirm RuneScape is restored when the sleep finishes.

23. Conversion Mappings

Older pattern	Current replacement
Loop, %runcount% + Begin/EndLoop	LLARS_RunCount(Func("Run"))
firstrun global	ctx.IsFirst
IniRead coordinates + Random + NaturalClick	LLARS_Click("Section")
IniRead hotkey + Send	LLARS_PressHotkey("Section")
Purposeful Send, {Space}	LLARS_PressKey("Space")
IniRead min/max + Random + Sleep	LLARS_Sleep("Section")
Final randomized sleep + LLARS_FinalSleep	LLARS_Sleep("Section", true)
Randomized/rescheduled label timer	LLARS_TimerRepeat("Section", Func("Callback"))
PixelGetColor + configured color read	LLARS_PixelMatches(...)
Several sequential color comparisons	LLARS_PixelMatchesAny(...)

24. Patterns to Avoid

- Do not use raw Send/Click/MouseClick for normal actions that have Creator API helpers.
- Do not keep creator-owned runcount/firstrun lifecycle in new RunCount scripts.
- Do not use dynamic section names for RunCount sleeps you expect the pre-run estimator to resolve.
- Do not override CoordMode in normal scripts. Core owns coordinate mode.
- Do not copy NaturalClick or other Core functions into a script.
- Do not duplicate SCRIPT_EDIT markers.
- Do not run raw game-facing actions without LLARS_WaitForRuneScape() immediately before them.
- Do not use Developer Mode as a dependency for script behavior.

25. Creator API Cheat Sheet

```
LLARS_RunCount(Func("Run"))
LLARS_SCRIPT_TYPE := "UntilDone"; LLARS_StartUntilDoneRun(); LLARS_EndUntilDoneRun()
LLARS_Click("Coordinate Section")
LLARS_Click("Coordinate Section", "right")
LLARS_PressKey("Space")
LLARS_PressHotkey("Hotkey Section")
LLARS_SetStatus("Working")

LLARS_Sleep("Sleep Section")
LLARS_Sleep("Final Sleep", true)
LLARS_RandomSleep()

TimerID := LLARS_TimerOnce("Timer Section", Func("Action"))
TimerID := LLARS_TimerRepeat("Timer Section", Func("Action"))
LLARS_TimerStop(TimerID)
LLARS_TimerStopAll()

color := LLARS_PixelColor("Pixel Coordinate")
matched := LLARS_PixelMatches("Pixel Coordinate", "Target Color")
which := LLARS_PixelMatchesAny("Pixel Coordinate", ColorSections)

value := LLARS_ConfigRead("Section", "key", "default")
flag := LLARS_ConfigReadBool("Feature")
count := LLARS_ConfigReadInteger("Settings", "count", "", 1)
number := LLARS_ConfigReadNumber("Settings", "value", 0)
text := LLARS_ConfigReadString("Settings", "name", "", true)
choice := LLARS_ConfigReadChoice("Settings", "mode", "One|Two")
LLARS_ConfigReadPoint("Location", x, y)
LLARS_ConfigReadRange("Timer", minValue, maxValue)

if !LLARS_WaitForRuneScape("Custom action")
    return
```

26. The Simple Version

- Use the current API for the normal thing first.
- Use Config.ini section names instead of repeating IniRead boilerplate.
- Use LLARS_RunCount(Func("Run")) instead of owning the loop.
- Use UntilDone start/end helpers when task logic determines completion instead of a timer or run count.
- Use LLARS_PressKey for purposeful fixed keys and LLARS_PressHotkey for user-configurable bindings.
- Use managed callback timers for script actions.
- Use LLARS_PixelMatchesAny for one pixel with several target colors.
- Use LLARS_WaitForRuneScape immediately before unavoidable raw game-facing code.
- Use the Legacy Guide when you intentionally need traditional AHK methods.